

Introduction

THERMAL ENGINEERING

Thermal Engineering deals with movements of heat energy and transfer. The energy can be transferred between two medium or transferred into other form of energy. Thermodynamics relate heat, work, temperature & energy. It deals energy transfer from one place to other & one form to another. The heat is a form of energy corresponding to the definite amount of mechanical work.

Thermodynamic state

A thermodynamic state is a system's condition at any given time. These properties are characteristic parameters that have definite value at each state and independent to the way in which system arrived at state. Change in value of property depend only on Initial & final state of system not on the path followed from one state to another. Such property called function.

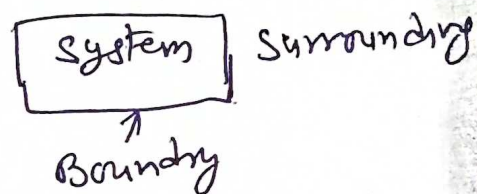
Thermodynamic Equilibrium

In which there is no tendency for the state of a system to change spontaneously, if the system is isolated from surroundings.

Basic Concept of Thermal Engineering

System:- A region in space containing the quantity of matter whose behaviour is being investigated

Everything external to the system is called surrounding. The system is separated by Boundary from surrounding



Closed system:- A system through which only energy (transfer) may cross the system Boundary but mass cannot transfer.

Exp. A quantity of fluid in the cylinder bounded by piston

Open system:- A system through which Both mass and energy may cross the system boundary.

Exp.:- A hot water in open Jar.

Isolated system:- A system through which neither energy nor mass crosses the system Boundary is called Isolated system.

Homogeneous and Heterogeneous System

A system consisting of single phase is called a homogeneous system while a system consisting more than one phase is known as a Heterogeneous system.

Mass:- A quantity of matter inside the body (m)

Weight:- The amount of pull through which the earth pull any body towards its center.

Weight = mass $\times$ gravity

$$W = m \times g. \text{ Newton}$$

force :- Anything that can change the position of object

$$F = ma \text{ By second law of Newton's}$$

Temperature :- The degree of hotness or coldness

$$\frac{C}{5} = \frac{F-32}{9} = \frac{F-273}{5}$$

Pressure :- force per unit area .

$$1 \text{ Bar} = 10^5 \text{ N/m}^2, \text{ In SI pascal unit} = 1 \text{ N/m}^2$$

Energy :- The capacity of doing work .

Kinetic Energy :- Energy of Body due to its motion

$$KE = \frac{1}{2}mv^2$$

Potential Energy :- The energy of Body due to its position above the ground is called potential Energy.

$$PE = mgh$$

Internal Energy :- (U) Energy of a body due to its molecular movement is called Internal energy

$$\text{Total Energy } E = PE + KE + U$$

Intensive property :- are those property of system which does not depend upon the mass as Temperature .

Extensive property :- properties which depend upon the mass of the body are called extensive property as Entropy

Laws of perfect Gas:-

- (1) Boyle's law :- At constant temperature, pressure of given mass of perfect gas is inversely proportional to its volume.

$$P \propto \frac{1}{V}$$

$$PV = \text{constant}$$

- (2) Charles's law :- At constant pressure for a perfect gas volume of given mass is directly proportional to its temperature.

$$V \propto T$$

$$\frac{V}{T} = \text{constant}$$

- (3) Gay Lussac's law :- At constant volume the pressure of given mass of perfect gas is directly proportional to its temperature.

$$P \propto T$$

$$\frac{P}{T} = \text{constant}$$

- (4) Joule's law :- For a perfect gas change in internal energy of a system is directly proportional to the change in temperature.

$$du \propto dt$$

$$du = m c_v dt \quad c_v = \text{specific heat at constant volume.}$$

The ideal gas equation

$$PV = nRT$$

$n = \text{No. of moles.}$

$$\text{or } PV = nMRT$$

$$PV = mRT$$

$R = \text{Universal gas constant } 8.314 \text{ J/mol}\cdot\text{K}$
 $= 259.8 \text{ Nm/kg}\cdot\text{K}$

$$R_0 = 8314.3 \text{ Nm/mol}\cdot\text{K}$$

Specific heat :- specific heat is defined as the amount of heat required to raise the temperature at 1°C of 1 kg substance.

C_p & C_v = Specific heat at constant pressure and volume.

Specific Heat at constant Volume C_v

The rate of change of Internal energy with respect to absolute temperature at constant volume (C_v)

$$C_v = \left(\frac{\partial u}{\partial T} \right)_v \quad u = \text{is internal energy} \\ T = \text{absolute Temp.}$$

$$\begin{aligned} \Phi &= \Delta u + W = \Delta u + P \Delta v \\ &= \int_{T_1}^{T_2} C_v dT \end{aligned}$$

Enthalpy is sum of internal energy and product of pressure and volume

$$h = u + PV$$

Specific Heat at constant Pressure (C_p)

The rate of change of enthalpy with respect to absolute temperature when pressure is constant - (C_p)

$$C_p = \left(\frac{\partial h}{\partial T} \right)_p$$

$$\Delta h = \Delta \Phi = \int_{T_1}^{T_2} C_p dT$$

$$C_p - C_v = \frac{R}{J}$$

$$\text{and } \gamma = \frac{C_p}{C_v}$$

Non flow process

Enthalpy H $\Delta H = \Delta E + P \Delta V$

Entropy S $\Delta S = \frac{\Delta Q}{T}$

Entropy is the amount of intrinsic disorder within the system (No exchange of mass)

Non flow Equation

$$Q = W + \Delta U$$

Non uniform flow: - a flow in which characteristic & parameter vary and different at different location.

(1) Constant Volume process

In this process volume remain constant i.e. $\Delta V = 0$ This also known as isochoric process By first law of thermodynamics

$$Q = \Delta U + W$$

$$W = \int P dV = 0$$

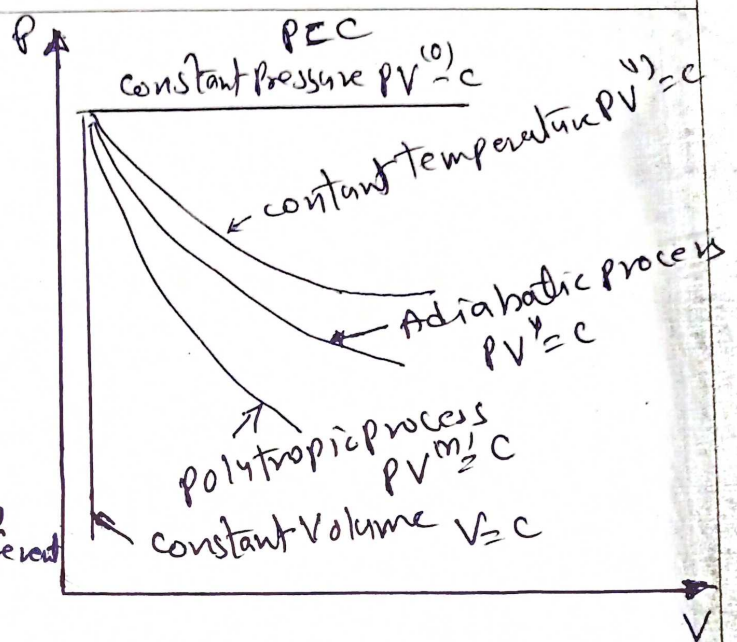
$$Q = \Delta U = m(u_2 - u_1) = m C_V dT \\ = m C_V (T_2 - T_1) \text{ KJ}$$

(2) constant Pressure (Isobaric process)

In this process pressure remain constant - $\Delta P = 0$

$$W = P dV = P (V_2 - V_1)$$

$$\int_1^2 dQ = \int_1^2 dU + \int_1^2 dW$$



$$\begin{aligned}
 Q_{1-2} &= m C_V (T_2 - T_1) + P (V_1 - V_2) \\
 &= m \frac{R}{\gamma - 1} (T_2 - T_1) + m R (T_2 - T_1) \\
 &= \frac{m R (T_2 - T_1) \gamma}{\gamma - 1}
 \end{aligned}$$

(3) constant Temperature process (Isothermal process)

$$PV = P_1 V_1 = P_2 V_2 = \text{constant}.$$

$$\begin{aligned}
 \text{Work } W_{1-2} &= \int_1^2 P dV = \int_1^2 \frac{P_1 V_1}{V} dV = P_1 V_1 \log_e \frac{V_2}{V_1} = P_1 V_1 \log \frac{P_1}{P_2} \\
 &= n R T \log_e \left(\frac{P_1}{P_2} \right)
 \end{aligned}$$

$$Q_{1-2} = W_{1-2} + (U_2 - U_1) = W_{1-2} + 0$$

$$U_2 - U_1 = m C_V (T_2 - T_1) = 0$$

(4) Adiabatic process (

Heat transfer is equal to zero

$$\text{workdone } W_{1-2} = \frac{P_1 V_1 - P_2 V_2}{\gamma - 1}$$

$$dQ = dU + dW$$

$$Q_{1-2} = (U_2 - U_1) + \frac{P_1 V_1 - P_2 V_2}{\gamma - 1}$$

$$0 = m C_V (T_2 - T_1) + \frac{P_1 V_1 - P_2 V_2}{\gamma - 1}$$

$$W_{1-2} = m C_V (T_2 - T_1)$$

(5) Polytropic process workdone $W_{1-2} = \frac{P_1 V_1 - P_2 V_2}{n - 1}$
 $P V^n = \text{constant}$

$$Q_{1-2} = m C_V (T_2 - T_1) \frac{(\gamma - n)}{1 - n}$$

Thermodynamic laws

(1) Zeroth law of thermodynamics :-

When two systems are each in thermal equilibrium with third system, the first two systems are in thermal equilibrium with each other.

(2) First law of thermodynamics

It is also known as conservation of energy. The change in a system's internal energy is equal to the difference between heat added to the system from surroundings and work done by the system on the surroundings.

The energy can neither be created nor be destroyed but can be changed from one form to another form.

i.e.

$$Q = \Delta U + W$$

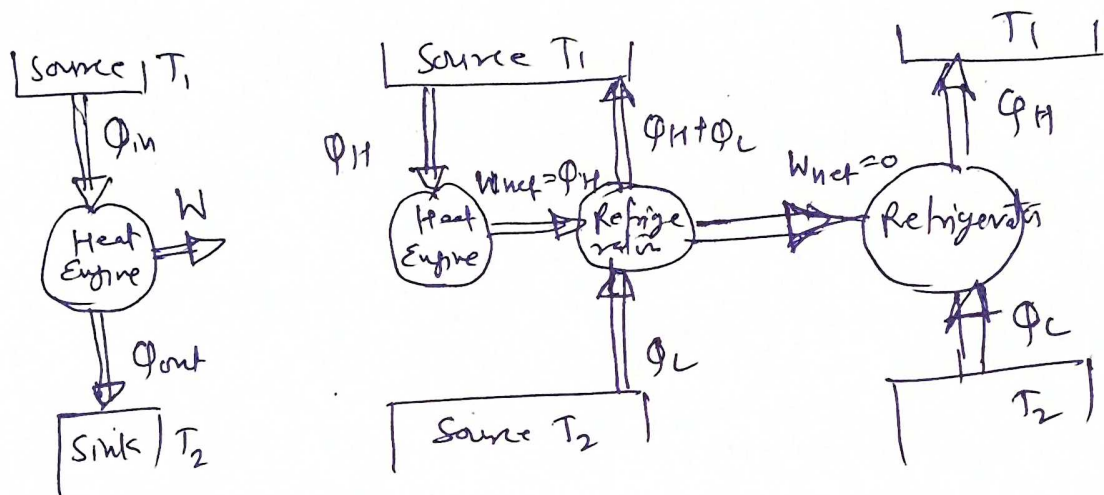
Q : amount of heat added ΔU : Increase in internal energy

W : Work done by the system

(3) Second law of thermodynamics

(a) Kelvin Planck's statement - It is not possible to convert all the heat extracted from a hot body (Q_H) into work (W), without losing some heat (Q_C) to a cold body.

(b) Clausius Statement - It is impossible to transfer heat from a hot body to a cold body without expenditure to work by external energy source.



Third law of thermodynamics

The entropy of perfect crystal of an element in its most stable state form tend to zero as the temperature approaches absolute zero. This allow an absolute scale for entropy to established that from a statistical point of view determine the degree of randomness or disorder in a system.

Applications of thermal engineering

- 1) Boiler design
- 2) Heat exchanger
- 3) Refrigeration system

(4) Solar heating

(5) Thermal Insulation

(6) Thermal power plants.

Heat Engine, Refrigerator and Heat Pump

Coefficient of Performance (COP) is defined as ratio of useful heating or cooling effect of to work required.

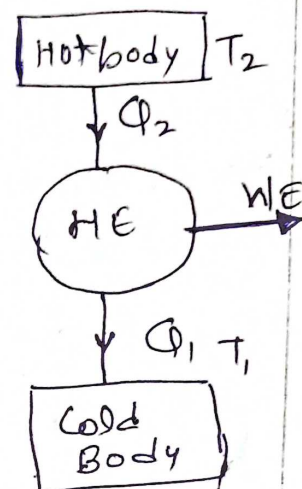
Heat Engine

C.O.P of Heat Engine is express in efficiency.

Heat Engine took Q_2 amount of ~~heat~~ heat from hot body and did work equivalent to W_E

The heat supplied to sink is equal to Q_1

Here useful effect is workdone W_E . It is equal to $Q_2 - Q_1$



C.O.P of heat engine, engine efficiency $\eta = \frac{\text{Workdone}}{\text{Heat supplied}}$

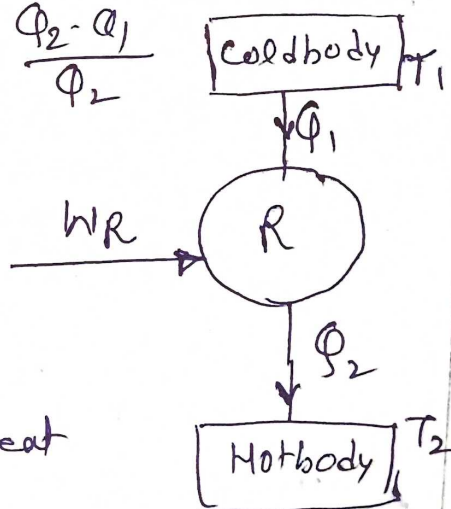
$$\eta = \frac{Q_2 - Q_1}{Q_2}$$

Refrigerator

In Refrigerator Q_1 amount of heat extracted from cold body and deliver ^{Q_2} to amount of heat to the hot body with the help of

Input work W_R . Here $Q_2 = Q_1 + W_R$

In Refrigerator useful effect is extraction of heat Q_1



Then COP of refrigeration is

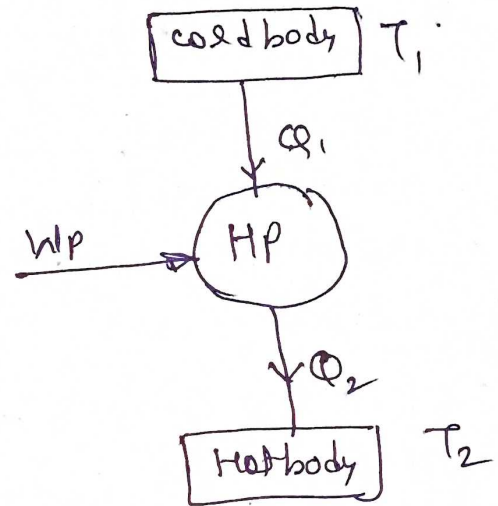
$$\begin{aligned} \text{COP}_R &= \frac{Q_1}{W_R} \\ &= \frac{Q_1}{Q_2 - Q_1} \end{aligned}$$

Heat pump

Heat pump is similar to refrigerator but in this case the desired effect is heat delivered to Hot body Q_2

The COP of heat pump is

$$\begin{aligned} (\text{COP})_p &= \frac{Q_2}{W_p} \\ &= \frac{Q_2}{Q_2 - Q_1} \end{aligned}$$

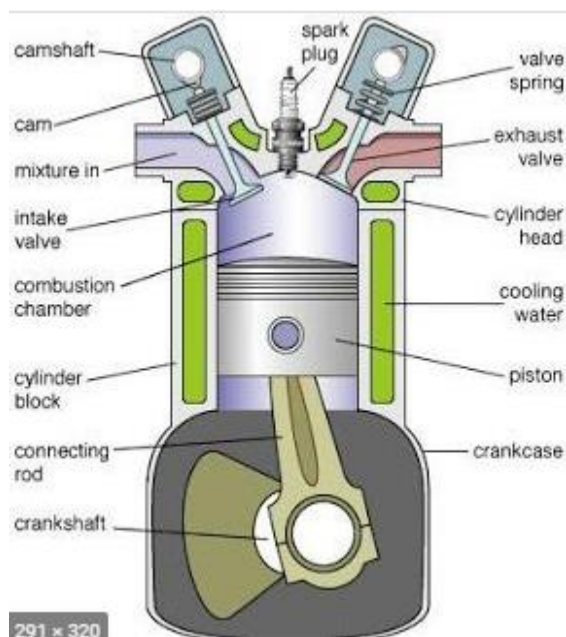


The relation between the COP of refrigerator and COP of heat pump.

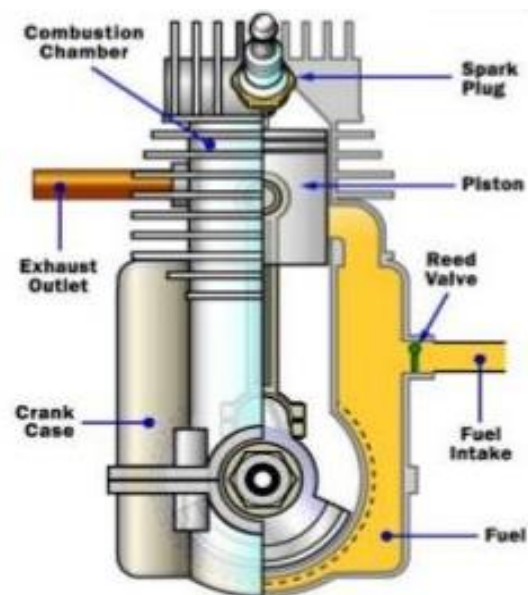
$$\begin{aligned} (\text{COP})_p &= \frac{Q_2}{Q_2 - Q_1} \\ &= 1 + \frac{Q_1}{Q_2 - Q_1} \end{aligned}$$

$$(\text{COP})_p = 1 + (\text{COP})_R$$

COP of heat pump is always greater than the unity



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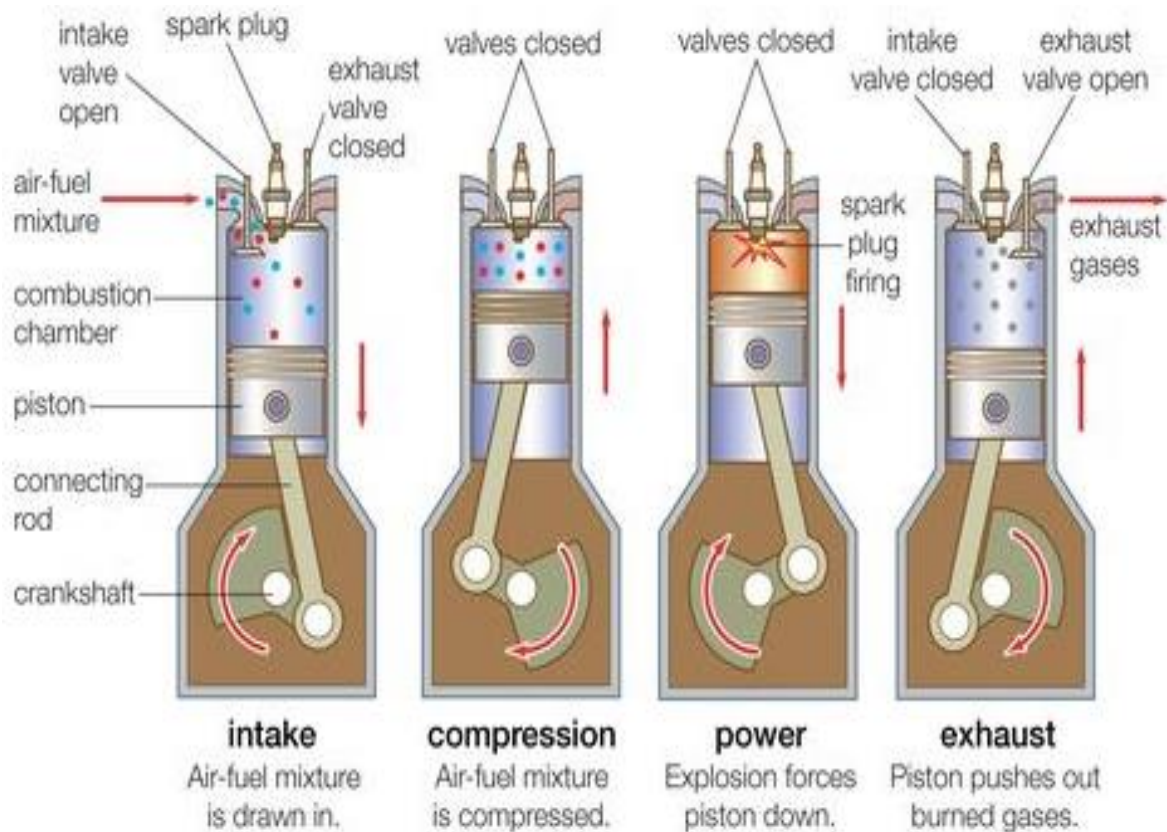
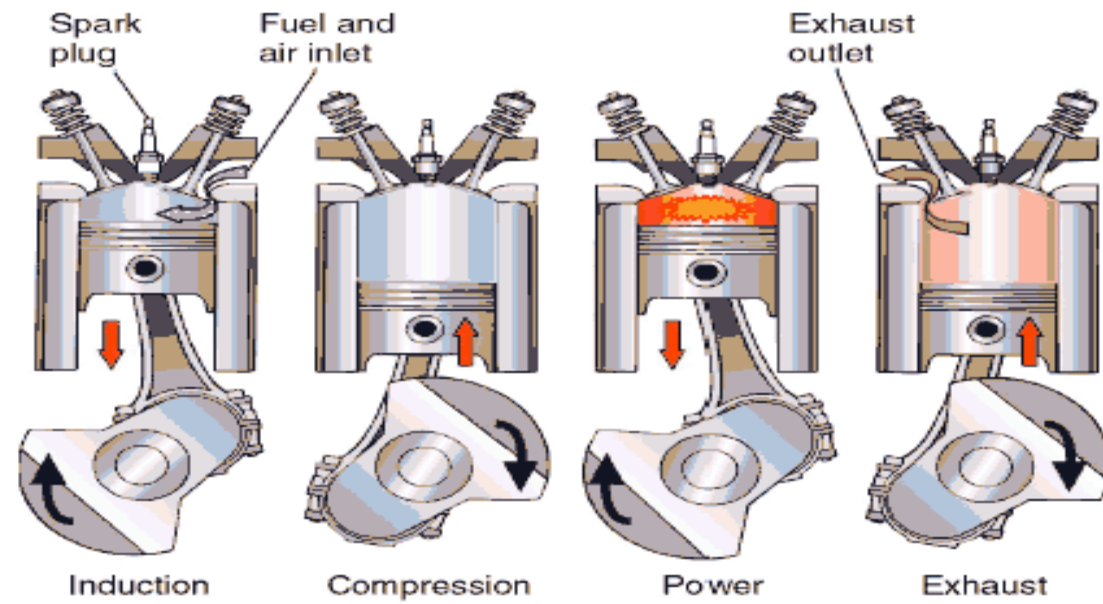


The basic components of a two-stroke engine

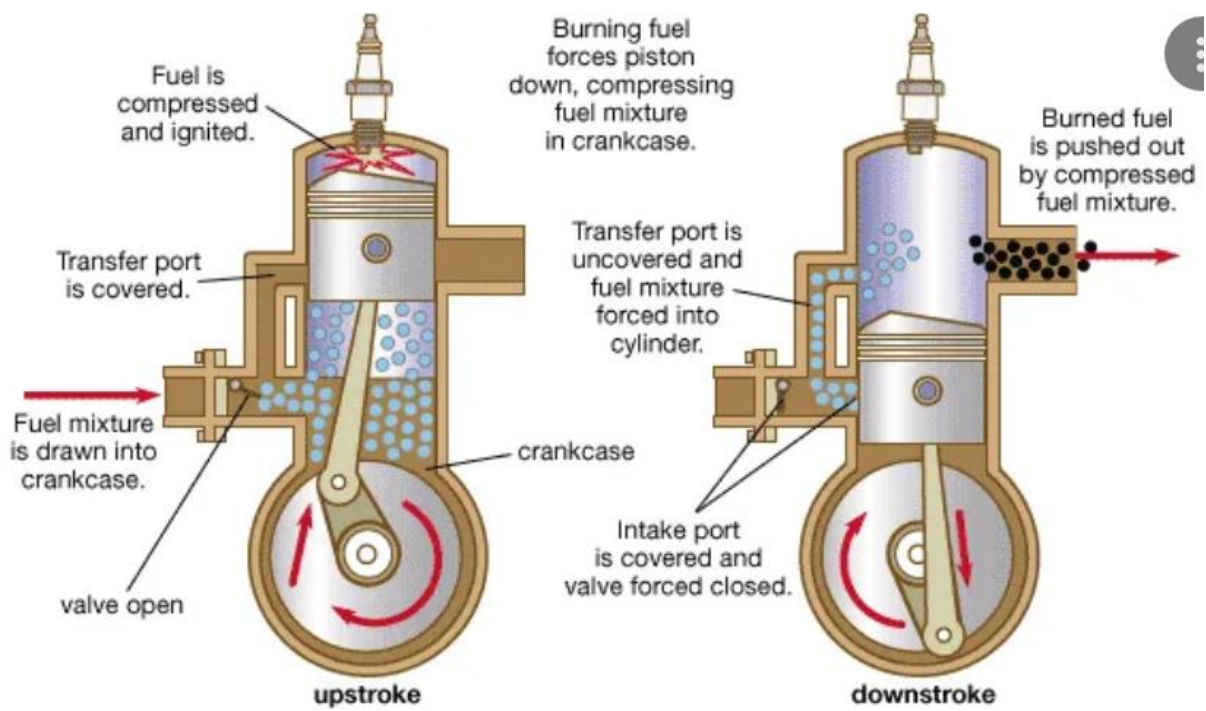
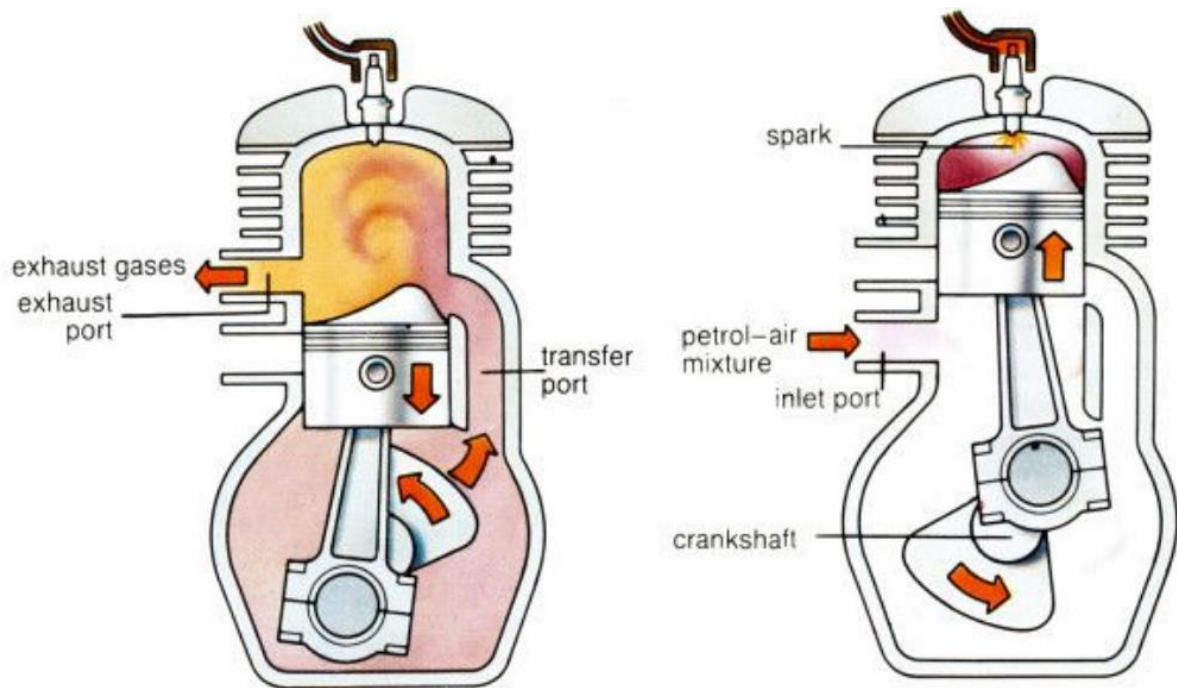
S. No.	Name of the Parts	Materials of Construction
1.	Cylinder head	Cast iron, Cast Aluminium
2.	Cylinder liner	Cast steel, Cast iron
3.	Engine block	Cast iron, Cast aluminum, Welded steel
4.	Piston	Cast iron, Aluminium alloy
5.	Piston Pin	Forged steel, Case hardened steel.
6.	Connecting rod	Forged steel. Aluminium alloy.
7.	Piston rings	Cast iron, Pressed steel alloy.
8.	Connecting rod bearings	Bronze, White metal.
9.	Main bearings	White metal, Steel backed Babbitt base.
10.	Crankshaft	Forged steel, Cast steel
11.	Camshaft	Forged steel, Cast iron, cast steel,
12.	Timing gears	Cast iron, Fiber, Steel forging.
13.	Push rods	Forged steel.
14.	Engine valves	Forged steel, Steel, alloy.
15.	Valve springs	Carbon spring steel.
16.	Mani folds	Cast iron, Cast Aluminium.
17.	Crankcase	Cast iron, Welded steel
18.	Flywheel	Cast iron.
19.	Studs and bolts	Carbon steel.
20.	Gaskets	Cork, Copper, Asbestos.

IC Engine parts

Four stroke Engine



Two stroke Engine



IC Engine

(11)

Internal Combustion Engine has extensively used in transport by road, rail, sea and air. IC engine have led to major revolution in transport

Classification of I.C. Engine: In following ways

(1) According to the type of fuel used

(1) petrol engine

(2) Diesel engine

(2) According to the cooling system

(1) Air cooled

(2) water cooled

(3) According to the cycle of operation

(1) four stroke cycle engine

(2) Two stroke cycle engine

(4) According to the charge pressure.

(1) Naturally aspirated engine

(2) Supercharged/turbocharged engine for higher capacity.

(5) According to the number of cylinders

(1) Single cylinder engine

(2) Multi cylinder engine

Main parts of I.C. Engine

(1) cylinder

The cylinder allow the piston to move to and fro. The cylinder is made of cast iron, steel or an aluminium alloy. Sometimes it made by cast-iron.

(2) cylinder Head.

It is fitted at the top of cylinder. A gasket is provided between cylinder and cylinder head to prevent the leakage of hot gases. The cylinder head also accommodates Inlet, Exhaust Valve, spark plug and injector etc.

(3) piston

It transmit the force exerted by burning gases to the connecting rod and finally to crank shaft. The piston is made of cast-iron or steel or aluminium alloy. The diameter of piston in a large engine is about 1 meter.

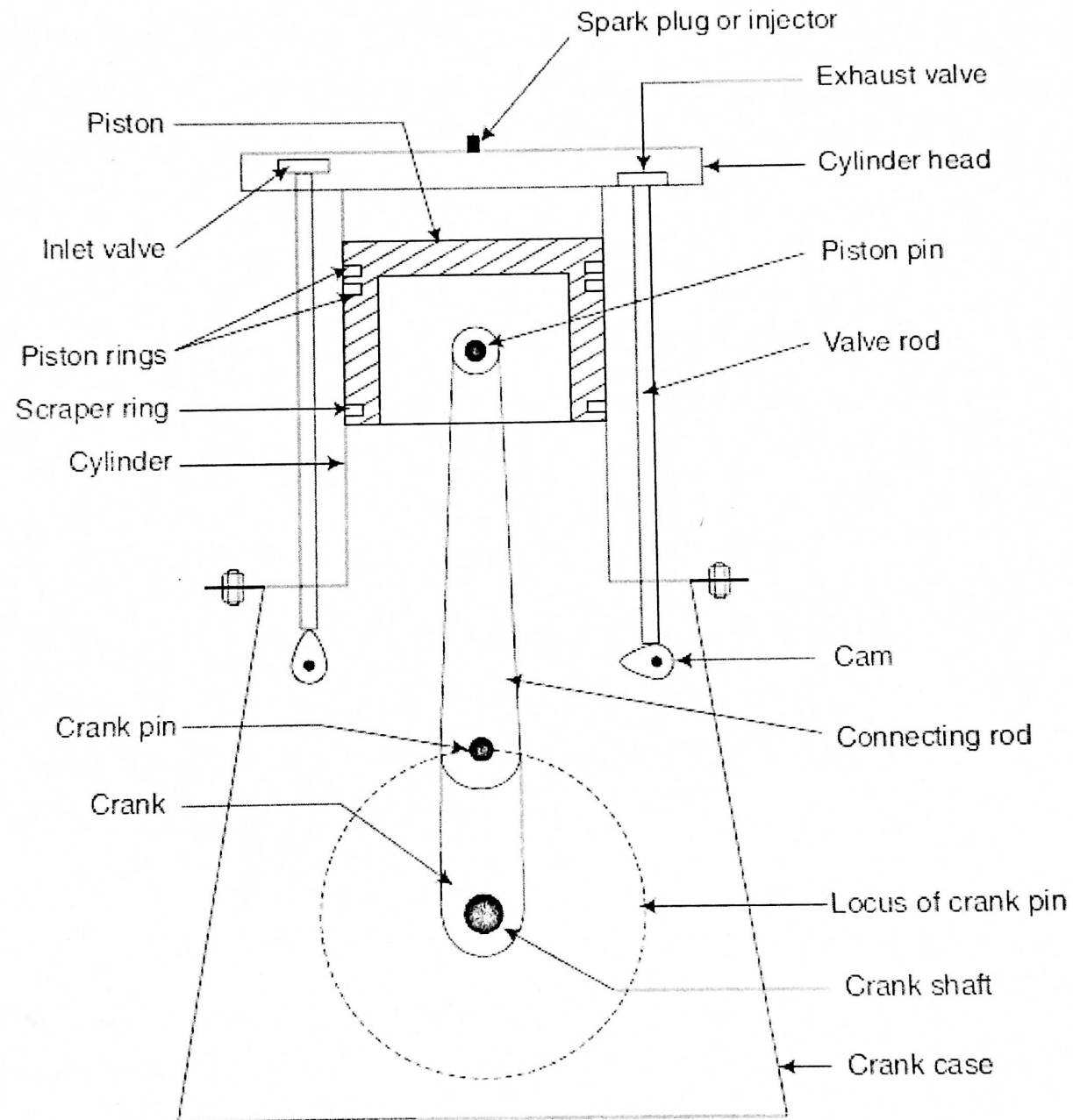
(4) piston ring.

Two different types of piston rings are houses in circumference grooves provided on the outer surface of ~~the~~ piston. The function of upper rings known as compression ring is provided gas tight sealing to maintain compression pressure inside the cylinder and to prevent the leakage of burnt gases into the crankshaft. The function of lower rings is to scrap the used lubricating oil into the crank case. This ring is called scraper ring or oil ring.

(5) connecting Rod.

This transmit the force from the piston to crankshaft

Main Parts of IC Engine



It is also helps to converting the reciprocating motion of the piston into the rotary motion of the crankshaft. For lubrication of piston pin in the connecting rod a small hole is provided from big end to small end. The small end of connecting rod is attached to the piston by a piston pin or gudgeon pin.

Crankshaft

Alloy steel are used to the crankshaft to withstand high stress and strain. The crankshaft is provided with suitable holes to help in lubrication system. The crank serve as a sump for lubricating oil.

Flywheel

It is mounted on the crankshaft. The flywheel store the excess energy during the power stroke of the engine and help the movement of the piston during the remaining idle strokes.

Cams

Properly designed cam control the opening and closing of the inlet and exhaust valve in the case of four stroke engine cam are rotated by a cam shaft driven by crank shaft through gear.

Compression Ratio

$$\text{Compression Ratio} = \frac{V_s + V_c}{V_c}$$

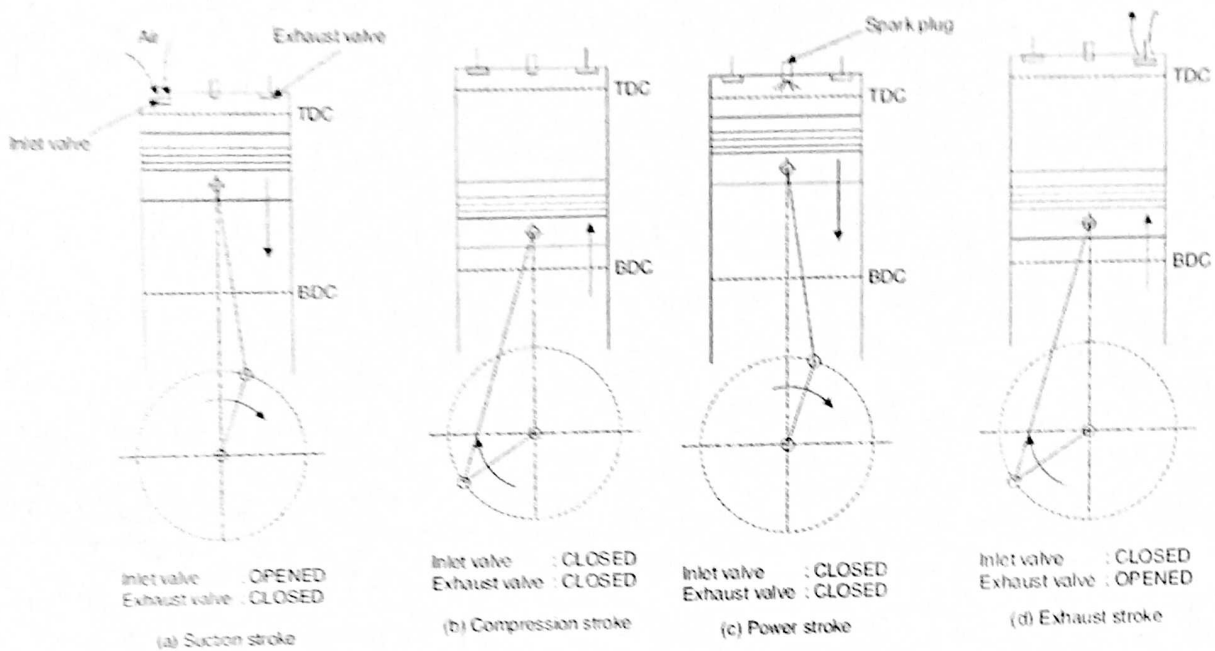
V_c = Clearance Volume

V_s = Swept Volume or Stroke Volume

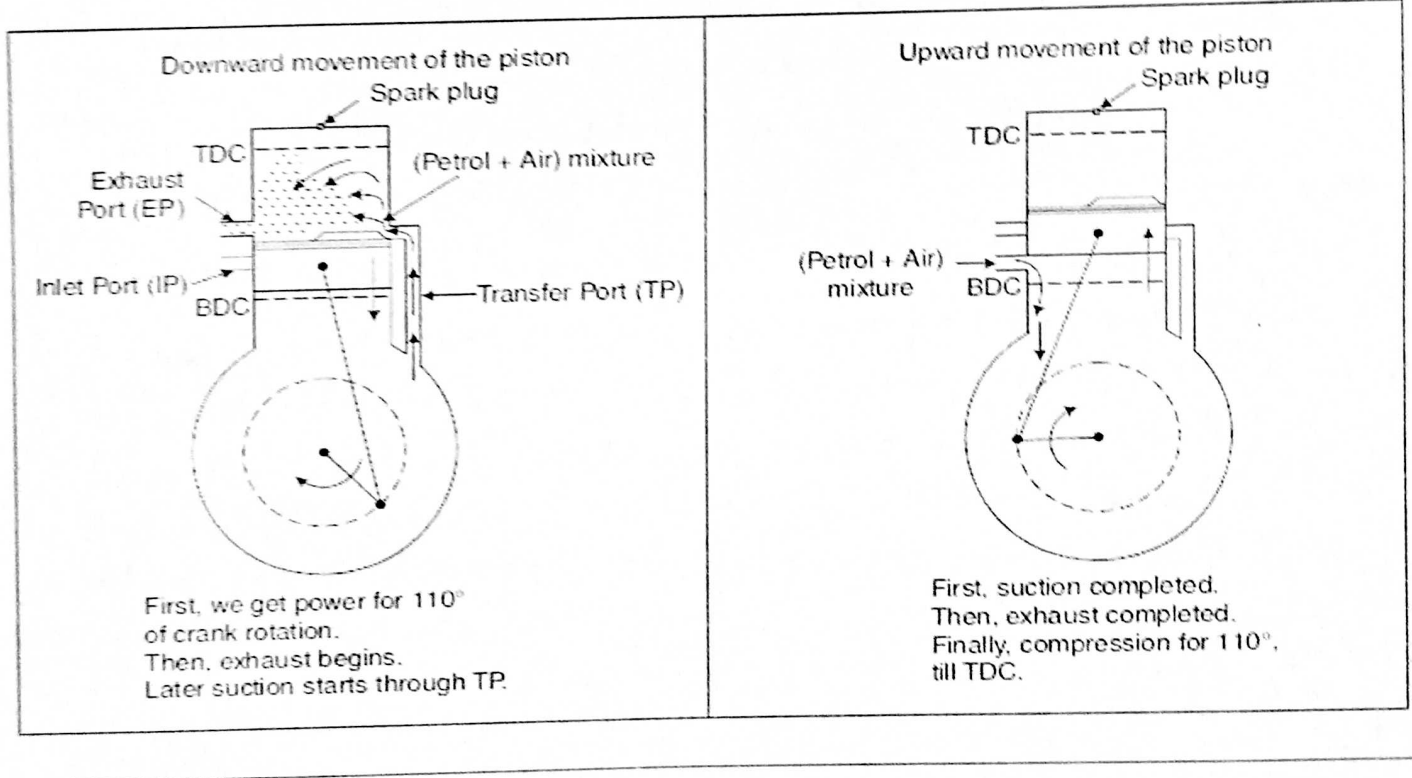
Working of Four Stroke petrol engine

- (1) Suction Stroke During this stroke the Inlet Valve is kept opened and exhaust valve is closed. The piston come down to the bottom dead centre (BDC) from the top dead centre (TDC). Pressure in the cylinder will be slightly less than the atmospheric pressure. petrol air mixture in correct proportion from carburetor is drawn inside the engine cylinder through inlet valve.
- (2) Compression Stroke In this both Inlet and exhaust valve closed. The air fuel mixture is compressed when piston moved up to TDC. The compression ratio varies from 7-10 for petrol engine. The ratio between the cylinder volume before compression to the volume after compression is called compression Ratio. At the end of Compression ~~stroke~~ stroke a spark is produced at a spark plug due to which combustion start resulting in high pressure and temperature which are comparatively less than that of a diesel engine.
- (3) Power Stroke During this stroke both valves are kept closed. The piston is pushed down from TDC to BDC. The force above the piston is transmitted to the crank shaft through the connecting rod and crank Mechanism. Excess energy due to combustion is stored in the flywheel which kept operation for other three idle stroke.

Four Stroke Engine



Two Stroke Engine



(4) Exhaust Stroke During the stroke the exhaust valve is kept opened and inlet valve is kept ~~closed~~. The piston moves up ~~and~~ from BDC to TDC. The waste gases are sent out through the exhaust valve and the cycle is repeated.

During these four strokes the crank shaft will make two revolutions. The thermal efficiency of a four stroke engine is higher compared to a two stroke cycle engine. Most of cars operates on a 4 stroke engine.

Four Stroke Diesel Engine

Diesel engine are also called compression ignition engine. There is no need for a spark plug.

(1) Suction Stroke :- During this stroke Inlet valve open and exhaust valve closed. The piston moved from TDC to BDC. Pressure in cylinder will be slightly less than the atmospheric pressure. Air is drawn inside the engine cylinder through inlet valve.

(2) Compression Stroke :- During this stroke Both inlet and exhaust valves ~~closed~~. The air is compressed when piston moved from BDC to TDC. The compression ratio varies from 15-20 for diesel engine. Due to high compression higher temperature is obtained in the range of 550°C . At this junction diesel oil at high pressure is injected inside

the hot compressed air in an atomised form. Due to the special shape of combustion chamber the mixing of fuel and air takes place rapidly and mixture is ignited due to the high compression ratio temperature without the help of spark plug. Hence the diesel engine is also called Compression Ignition engine (C I engine). Due to combustion very high pressure is produced.

Power Stroke :- During this both inlet and exhaust ~~stroke~~ valve is kept closed. The piston is pushed down from TDC to BDC. The force above piston is transmitted to the crank shaft through connecting rod and crank mechanism. Excess energy due to combustion is stored in flywheel which helps the operation of three idle strokes.

Exhaust Stroke :- During this stroke the both valves are in different position exhaust valve open and closed the Inlet Valve. The piston moved from BDC to TDC the waste gases are sent through the exhaust Valve and cycle is repeated.

During these four strokes the crankshaft will make two revolution. The thermal efficiency of a 4 stroke engine is higher compared to 2 stroke cycle engine. At present most of cars operated on 4 stroke cycle.

Difference Between Petrol and Diesel Engine

	Petrol Engine	Diesel Engine
(1)	Compression Ratio 7-10	(1) Compression Ratio 15-20
(2)	Petrol air mixture is compressed	(2) air is compressed
(3)	compression pressure 15-20 bar	(3) compression pressure 30-40 bar
(4)	Thermal efficiency low in 20-25%.	(4) Thermal efficiency is 25-30%.

S No	Petrol Engine	Diesel Engine
(5)	Spark plug needed	(5) No Need of Spark plug
(6)	Due to low pressure thickness of parts is less	(6) Due to high pressure thickness of parts is more.
(7)	weight of engine is less	(7) weight of engine is more.
(8)	Cost of engine is less	(8) Cost of engine is more
(9)	Operating cost per km is high due to high cost of petrol	(9) Operating cost per km is comparatively low due to low cost

Two stroke engine.

- (1) Petrol - two stroke petrol engine
- (2) Diesel - two stroke Diesel engine.

Cooling system of engine

- (1) water cooling system
- (2) Air cooling system

Lubricating system

Types of lubricants

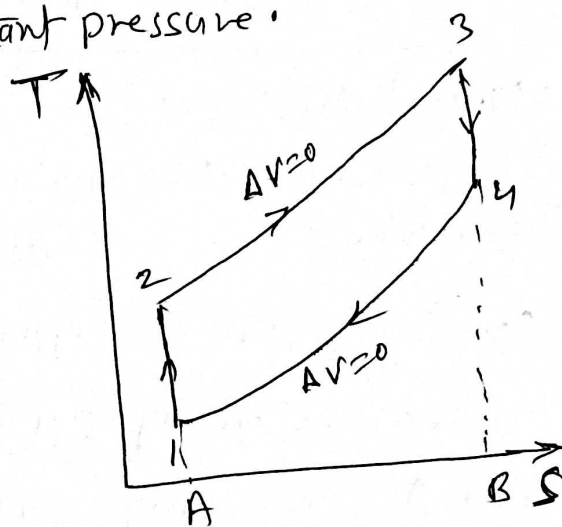
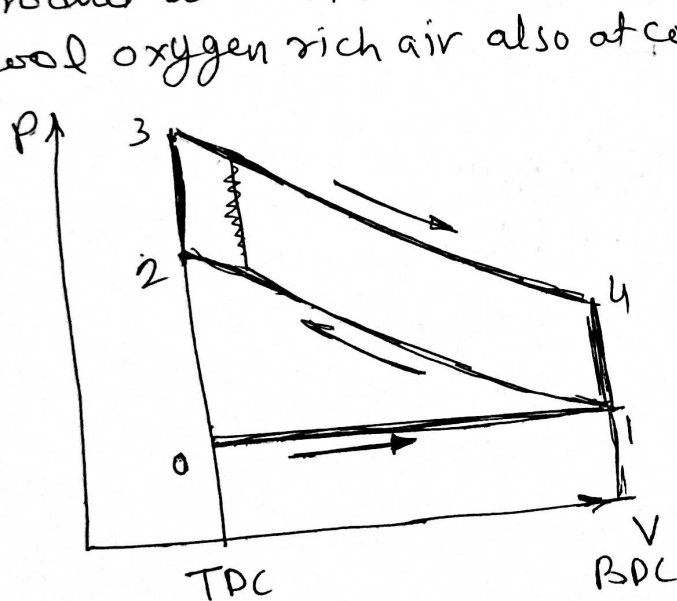
- (1) liquid like mineral oil vegetable oil etc
- (2) Semiliquid like grease
- (3) Solid lubricants like graphite powder with oil or grease.

Types of lubrication system

- (1) Splash lubrication system
 - (2) Pressure lubrication
 - (3) Petrol lubrication
- Working cycle

Otto cycle

It is thermodynamic cycle that describes the function of spark ignition piston engine. The Otto cycle consists of isentropic compression, heat addition at constant volume, isentropic expansion and rejection of heat at constant volume. In the case of four stroke Otto cycle there are two additional processes one for the exhaust of waste heat and combustion product at constant pressure (isobaric) and one for intake of cool oxygen rich air also at constant pressure.



- (1) Process 0-1
A mass of air is drawn into cylinder at constant pressure.
- (2) Process 1-2
It is an adiabatic (isentropic) compression of the charge as the piston moves from BDC to TDC.
- (3) Process 2-3
It is a constant volume heat transfer to the working

- gas from an external source while piston moves from TDC. The process is intended to represent the ignition of the airfuel mixture and subsequent rapid burning

(4) process 3-4

It is an adiabatic (isentropic) expansion power stroke

(5) process 4-1

Complete the cycle by a constant volume process in which heat is rejected from the air which piston is at BDC

(6) process 1-0

The mass is released to the atmosphere in a constant pressure process

Efficiency of Otto cycle

In the process from 1 to 2 and 3 to 4 are isentropic

$$T_4 V_1^{\gamma-1} = T_3 V_2^{\gamma-1}, \quad T_1 V_1^{\gamma-1} = T_2 V_2^{\gamma-1}$$

$$(T_4 - T_1) V_1^{\gamma-1} = (T_3 - T_2) V_2^{\gamma-1}$$

$$\frac{T_4 - T_1}{T_3 - T_2} = \left(\frac{V_2}{V_1} \right)^{\gamma-1}$$

$$= \left(\frac{1}{r} \right)^{\gamma-1}$$

$$\frac{V_1}{V_2} = \frac{V_1}{V_2} = r \text{ Compression ratio}$$

$$\frac{V_1}{V_2} = r$$

$$\text{Efficiency } \eta = \frac{\text{work}}{\text{heat input}}$$

$$\eta = \frac{\Phi_H + \Phi_L}{\Phi_H} = 1 + \frac{\Phi_L}{\Phi_H}$$

$$\text{Heat absorbed} = C_V(T_3 - T_2)$$

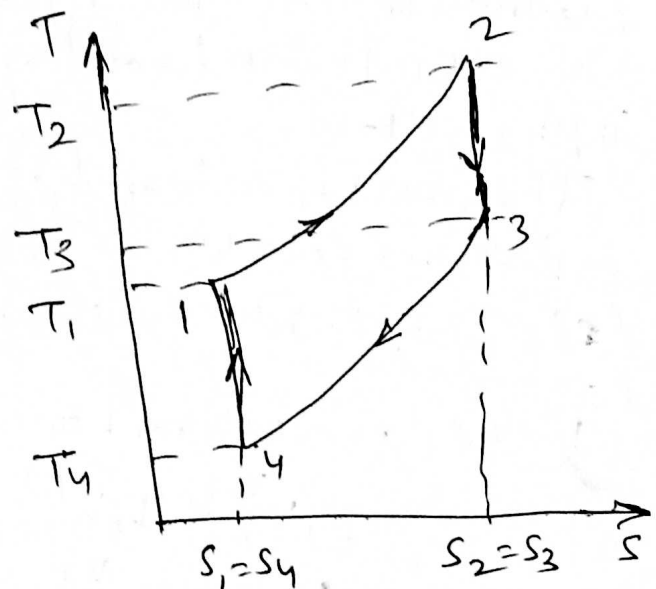
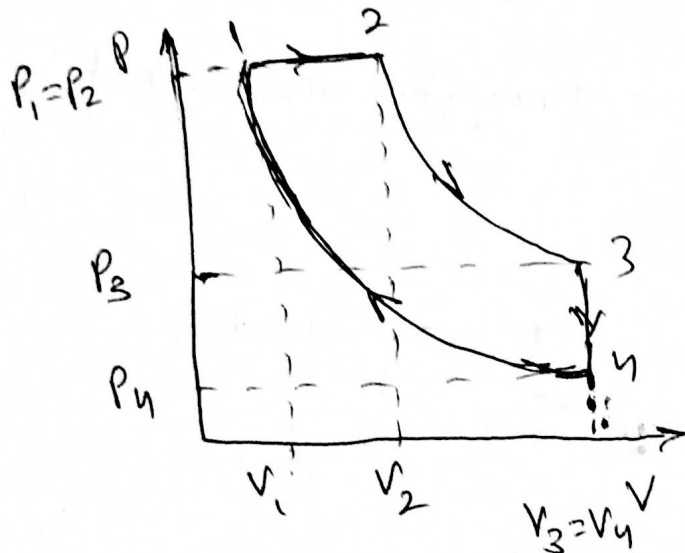
$$\text{Heat rejected} = C_V(T_1 - T_4)$$

$$\text{Hence } \eta = 1 - \frac{T_4 - T_1}{T_3 - T_2}$$

$$\eta = 1 - \frac{1}{r^{\gamma-1}}$$

The heat exchange is positive if heat is flow into the system
so Φ_L is negative.

Diesel Cycle



① Process 1-2 (Constant pressure heating)

The air is heated at constant pressure from initial Temp. T_1 to temp T_2 . The heat is cut off at point 2 so called cutoff point.

$$\text{Heat supplied to air} = m C_p (T_2 - T_1)$$

② Process (2-3) Adiabatic Expansion

The air is adiabatically expanded from T_2 temp to T_3 temperature.

(3) constant Volume cooling Process 3-4

The air is now cooled at constant volume from temp. T_3 to a temp T_4

$$\text{Heat rejected by the air} = m c_v (T_3 - T_4)$$

(4) Process 4-1 (Adiabatic Compression)

The air is compressed ~~adiabatically~~ from temp. T_4 to T_1

$$\text{work done} = \text{Heat supplied} - \text{Heat rejected}$$

$$= m c_p (T_2 - T_1) - m c_v (T_3 - T_4)$$

$$\text{Air standard efficiency} = \eta = \frac{\text{work done}}{\text{Heat supplied}}$$

$$= \frac{m c_p (T_2 - T_1) - m c_v (T_3 - T_4)}{m c_p (T_2 - T_1)}$$

$$= 1 - \frac{c_v (T_3 - T_4)}{c_p (T_2 - T_1)}$$

$$= 1 - \frac{1}{\gamma} \frac{(T_3 - T_4)}{T_2 - T_1}$$

$$\text{Compression ratio } r = \frac{V_4}{V_1}$$

$$\text{cut off ratio } \rho = \frac{V_2}{V_1}$$

Expansion Ratio $r_1 = \frac{V_3}{V_2} = \frac{V_4}{V_2} \because V_3 = V_4$

$$= \frac{V_4}{V_1} \times \frac{V_1}{V_2} = \frac{V_4}{V_2}$$

$$= r \times \frac{1}{e} = \frac{r}{e}$$

constant pressure heating process (1-2)

$$= \frac{V_1}{T_1} = \frac{V_2}{T_2} \text{ (Charls law)}$$

$$\therefore T_2 = T_1 \times \frac{V_2}{V_1} = T_1 \times e$$

Similarly adiabatic compression process (2-3)

$$\frac{T_3}{T_2} = \left(\frac{V_2}{V_3} \right)^{\gamma-1} = \left(\frac{1}{r_1} \right)^{\gamma-1} = \left(\frac{e}{r} \right)^{\gamma-1} \quad r_1 = \frac{r}{e}$$

In a adiabatic compression process (4-1)

$$\frac{T_1}{T_4} = \left(\frac{V_4}{V_1} \right)^{\gamma-1} = (r)^{\gamma-1}$$

$$T_1 = T_4 (r)^{\gamma-1}$$

substitute the value of T_1 we get

$$T_2 = T_4 (r)^{\gamma-1} \times e$$

$$T_3 = T_4 (r)^{\gamma-1} \times e \left(\frac{e}{r} \right)^{\gamma-1} = T_4 e^{\gamma}$$

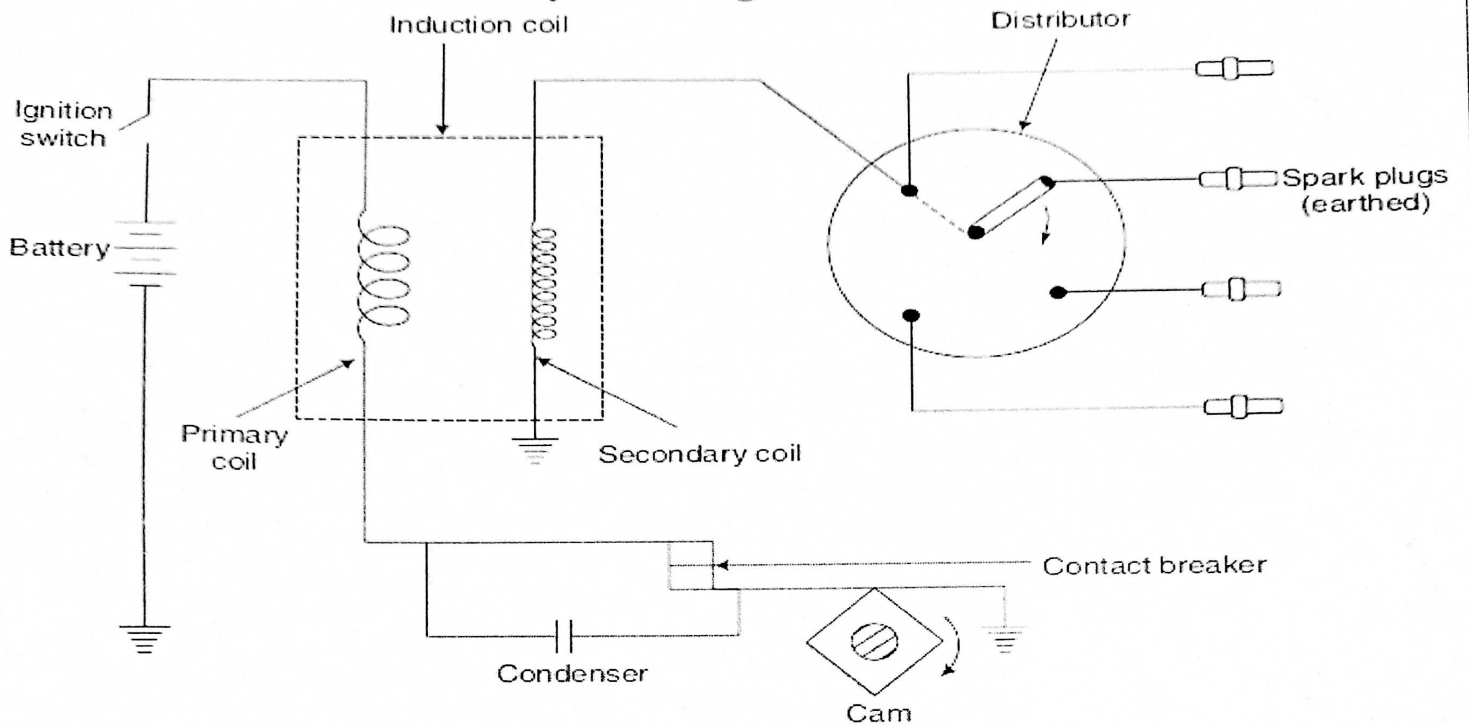
Now substitute the value of T_1 , T_2 and T_3 we get

$$\eta = 1 - \frac{1}{\gamma} \left[\frac{T_4 e^{\gamma} - T_4}{T_4 (r)^{\gamma-1} - T_4 (r)^{\gamma-1}} \right]$$

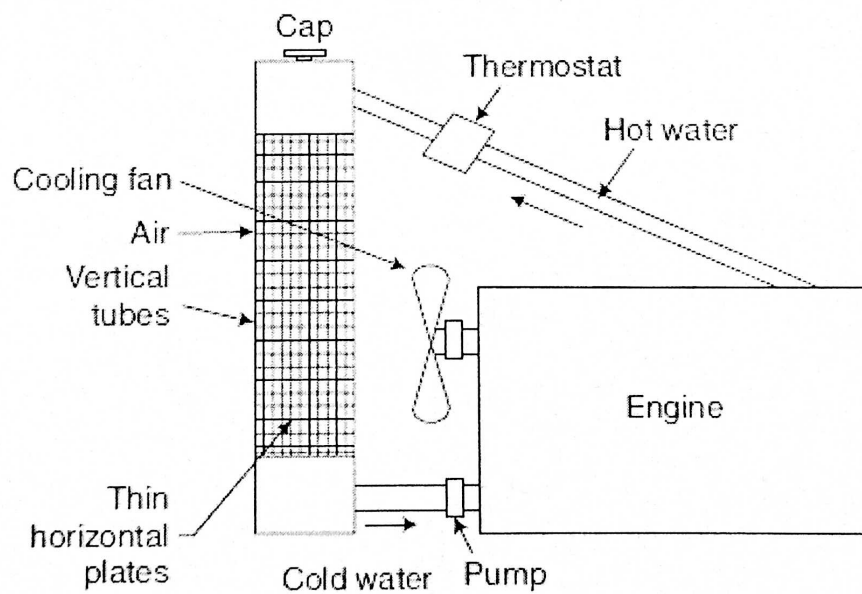
$$= 1 - \frac{1}{\gamma-1} \left[\frac{e^{\gamma} - 1}{\gamma(e-1)} \right]$$

Battery or Coil Ignition system

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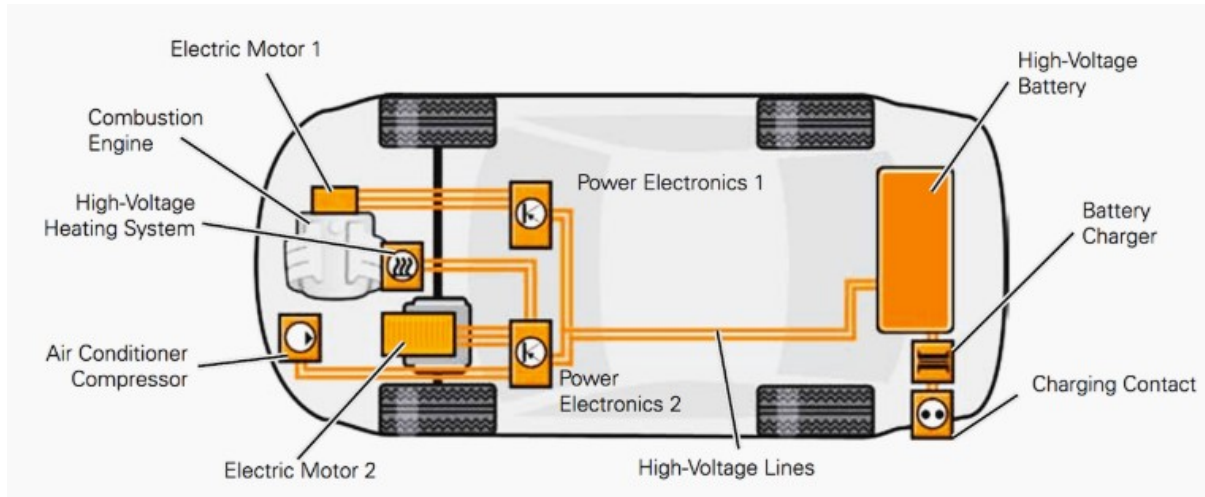


Cooling System



Basics of Electric Vehicle

The fundamentals of electric vehicles (EVs) involve using electricity from a battery to power an electric motor instead of a gasoline engine. Key components include a high-voltage battery pack for energy storage, an electric motor for propulsion, and a controller to manage the flow of electricity. EVs are charged by plugging into an external power source or through [regenerative braking](#), which captures energy during deceleration.

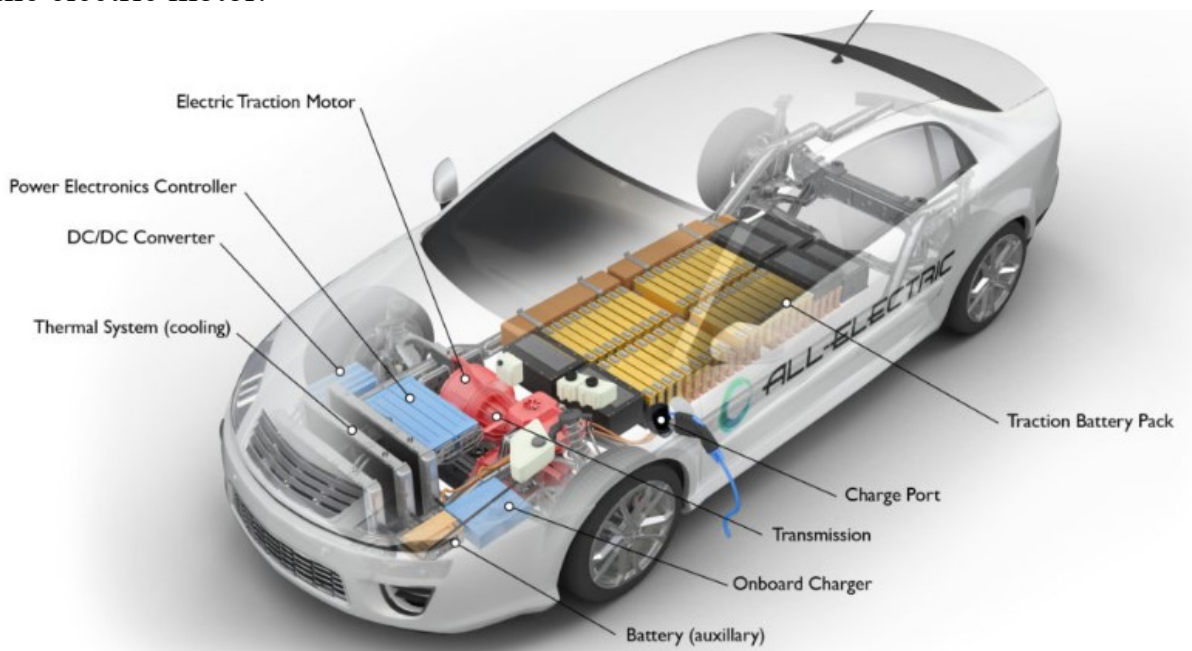


Key Components

While specific components vary by type, the main components in a BEV (the most common type of EV) are:

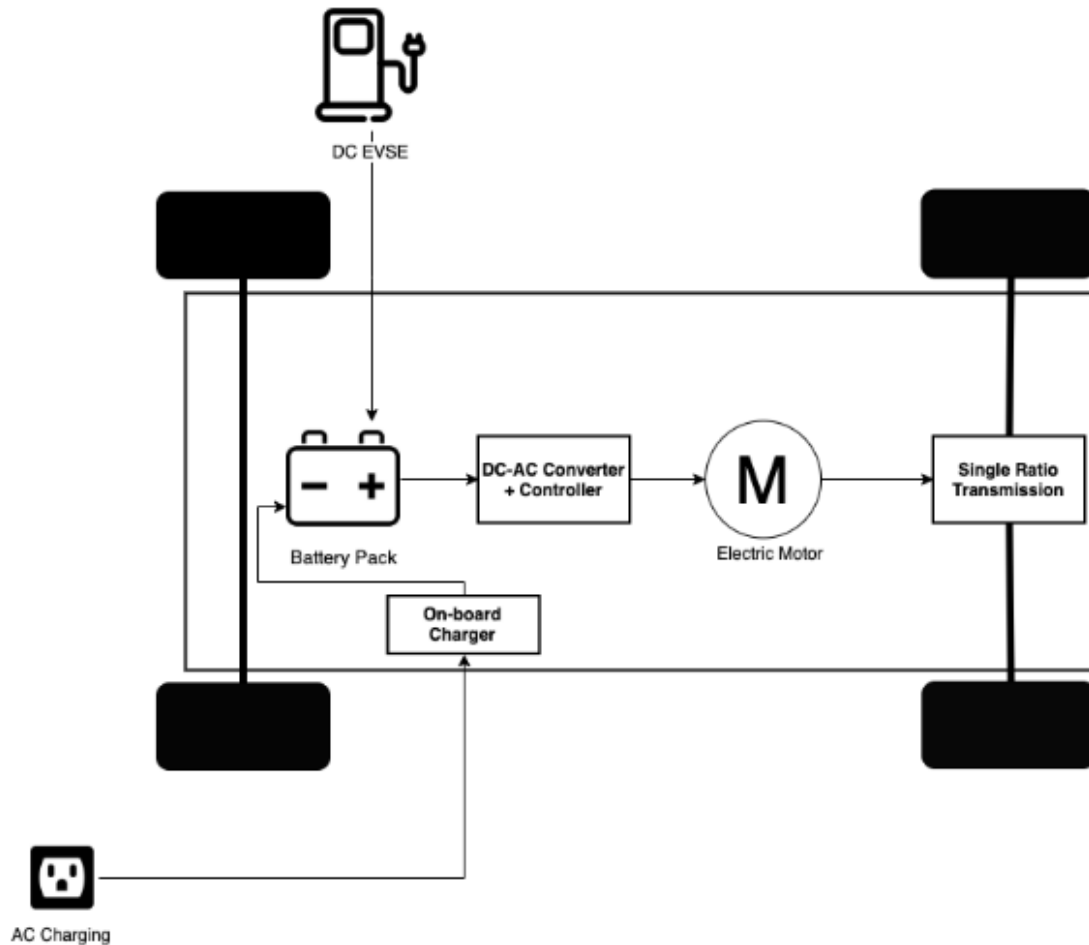
- **Traction Battery Pack:** The primary energy storage system, typically lithium-ion, that powers the electric motor.
- **Electric Traction Motor:** Converts electrical energy from the battery into mechanical energy to drive the wheels. These motors offer instant torque and high efficiency.
- **Onboard Charger:** Converts the AC electricity from an external charging source into DC power to charge the battery pack safely and efficiently.
- **Charge Port:** The inlet where the vehicle is plugged into an external power supply.
- **Power Electronics Controller:** Manages the flow of electrical energy, controlling vehicle speed and the torque produced by the motor.
- **Thermal System:** Maintains the optimal operating temperature for the battery, motor, and other electronic components to ensure safety and peak performance.

- **Regenerative Braking System:** A feature that captures kinetic energy during braking and converts it back into electricity to recharge the battery, which also reduces wear on brake pads.
- **Controller:**
Manages the flow of electricity from the battery to the motor. It receives signals from the accelerator and adjusts the motor's speed by changing the power frequency.
- **Inverter:**
Converts the direct current (DC) from the battery to alternating current (AC) for the electric motor.



How they work

- When the accelerator is pressed, the controller sends a signal to the inverter, which converts the battery's DC power to AC power for the motor.
- The motor then uses this electrical energy to produce mechanical force, turning the wheels.
- During braking or deceleration, the motor acts as an alternator, generating electricity that is sent back to the battery in a process called regenerative braking.



- **Charging**

EVs can be charged at home using a standard wall outlet or a dedicated home charger at workplaces, or at public charging stations. Public chargers range in speed, from slower AC Level 2 chargers to rapid DC fast chargers, which can provide a substantial charge in 20-30 minutes. Drivers can use tools like [PlugShare](#) or [EVgo](#) to locate nearby charging stations.

Types of electric vehicles

- **Battery Electric Vehicles (BEVs):**

Run entirely on electricity stored in their batteries. They have no combustion engine and produce zero tailpipe emissions.

- **Plug-in Hybrid Electric Vehicles (PHEVs):**

Combine a combustion engine with an electric motor and a battery that can be plugged in to charge. They can run on electricity alone for a moderate distance before the gasoline engine takes over.

- **Hybrid Electric Vehicles (HEVs):**

Use both a combustion engine and an electric motor, but their batteries are smaller and not plugged in; they are charged by the engine or through regenerative braking.

Benefits

- **Environmentally Friendly:** BEVs produce zero tailpipe emissions, which helps reduce air pollution and greenhouse gases.
- **High Efficiency:** Electric motors convert over 85% of electrical energy into motion, compared to less than 40% for gasoline engines.
- **Lower Operating Costs:** Electricity is often cheaper per mile than gasoline, and EVs have fewer moving parts, resulting in less maintenance.
- **Performance:** EVs provide instant torque and smooth, quiet acceleration due to the nature of electric motors.

Hybrid Vehicle

A hybrid vehicle is a car that combines a traditional internal combustion engine with an electric motor and battery to improve fuel efficiency and reduce emissions. The battery is recharged through regenerative braking and the engine itself, so most standard hybrids do not need to be plugged in. This setup allows for a seamless transition between gas and electric power, often resulting in lower fuel costs.

How it works

- **Power source:** The vehicle uses both a gasoline engine and one or more electric motors.
- **Battery:** An internal battery stores energy, and is recharged by a generator powered by the gasoline engine and by capturing energy that is normally lost during braking ([regenerative braking](#)).
- **Efficiency:** The electric motor can assist the engine, allowing for a smaller, more efficient engine. It can also power the vehicle at low speeds or in stop-and-go traffic, and can reduce engine idling when stopped.
- **Types:** Hybrids can be **full hybrids**, where the electric motor can power the car alone, or **mild hybrids**, where the electric motor only assists the engine.

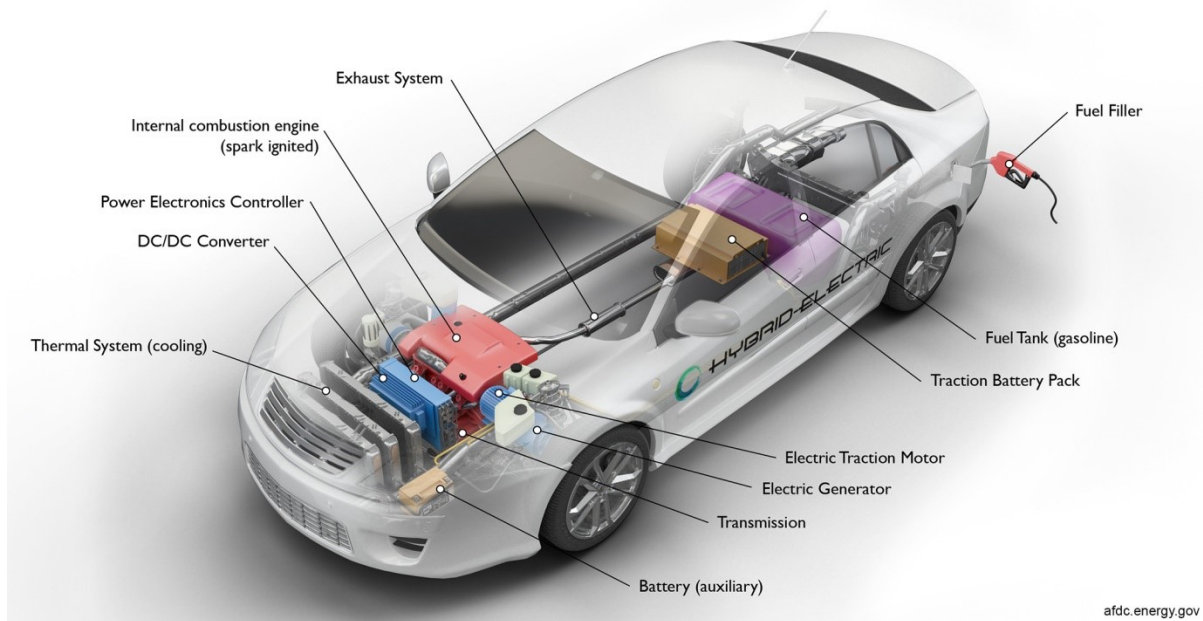
• Advantages and Disadvantages

Feature	Advantages	Disadvantages
Fuel Efficiency	Significantly better fuel economy, especially in stop-and-go city driving.	Fuel savings may be less significant on continuous highway driving.
Environmental Impact	Lower tailpipe emissions and a reduced carbon footprint compared to conventional vehicles.	Still produce some emissions, unlike pure electric vehicles.
Convenience	No range anxiety for non-PHEVs; refuel at any gas station. PHEVs offer flexibility for electric daily commutes and gas for long trips.	PHEVs require access to charging infrastructure to maximize electric benefits.
Cost & Maintenance	Less wear on friction brakes due to regenerative braking.	Higher initial purchase price than non-hybrid counterparts. Maintenance can be more complex and potentially costly if specialized hybrid components (like the battery pack out of warranty) need replacement.

Popular Hybrid Vehicle Models (2025)

- **Maruti Suzuki Grand Vitara:** A popular and affordable strong hybrid SUV known for its balance of price, mileage, and features.
- **Toyota Urban Cruiser Hyryder:** Built on the same platform as the Grand Vitara, offering similar strong hybrid performance and efficiency with a different design.
- **Honda City e:HEV:** A refined hybrid sedan that is praised for its excellent city mileage and smooth performance.
- **Toyota Camry Hybrid:** A benchmark for reliability and comfort in the luxury sedan segment, known for a quiet cabin and proven efficiency.
- **Volvo XC60 Recharge:** A luxury plug-in hybrid SUV that offers strong performance and a substantial electric-only range.

Hybrid Electric Vehicle



Key Components of a Hybrid Electric Car

Battery (auxiliary): In an electric drive vehicle, the low-voltage auxiliary battery provides electricity to start the car before the traction battery is engaged; it also powers vehicle accessories.

DC/DC converter: This device converts higher-voltage DC power from the traction battery pack to the lower-voltage DC power needed to run vehicle accessories and recharge the auxiliary battery.

Electric generator: Generates electricity from the rotating wheels while braking, transferring that energy back to the traction battery pack. Some vehicles use motor generators that perform both the drive and regeneration functions.

Electric traction motor: Using power from the traction battery pack, this motor drives the vehicle's wheels. Some vehicles use motor generators that perform both the drive and regeneration functions.

Exhaust system: The exhaust system channels the exhaust gases from the engine out through the tailpipe. A three-way catalyst is designed to reduce engine-out emissions within the exhaust system.

Fuel filler: A nozzle from a fuel dispenser attaches to the receptacle on the vehicle to fill the tank.

Fuel tank (gasoline): This tank stores gasoline on board the vehicle until it's needed by the engine.

Internal combustion engine (spark-ignited): In this configuration, fuel is injected into either the intake manifold or the combustion chamber, where it is combined with air, and the air/fuel mixture is ignited by the spark from a spark plug.

Power electronics controller: This unit manages the flow of electrical energy delivered by the traction battery, controlling the speed of the electric traction motor and the torque it produces.

Thermal system (cooling): This system maintains a proper operating temperature range of the engine, electric motor, power electronics, and other components.

Traction battery pack: Stores electricity for use by the electric traction motor.

Transmission: The transmission transfers mechanical power from the engine and/or electric traction motor to drive the wheels.

Benefits of hybrid vehicles

- **Improved fuel economy:** Hybrids are often more fuel-efficient, especially in city driving.
- **Lower emissions:** They produce fewer tailpipe emissions, which reduces their carbon footprint.
- **No range anxiety:** Unlike some electric-only vehicles, most hybrids don't require external charging, so you don't have to worry about running out of battery power as much.
- **Cost savings:** Over time, fuel savings can help offset the higher initial cost of the vehicle.
- **Higher resale value:** Due to advanced technology and increasing popularity, many hybrids have a high resale value.

Types of Hybrid Vehicles

Hybrid vehicles are categorized primarily by the degree of electric assistance and how the two power sources are connected.

- **Mild Hybrid (MHEV)**
 - **Function:** The small electric motor cannot power the car alone; it primarily assists the engine during acceleration and enables a smooth start-stop function to save fuel when idling.
 - **Charging:** The small battery is charged solely through regenerative braking and the internal combustion engine. No external charging is needed.
 - **Examples:** Some versions of the Land Rover Defender and Volkswagen Tiguan.

Full Hybrid (HEV or Strong Hybrid)

Function: The electric motor can run the vehicle independently for short distances and at low speeds, such as in city traffic. The system automatically switches between the engine, motor, or a combination of both for optimal efficiency.

Charging: The battery is "self-charging" via regenerative braking and the gas engine; it does not require plugging in.

Examples: The Toyota Prius, Toyota Camry Hybrid, Honda City e:HEV, Maruti Suzuki Grand Vitara, and Toyota Urban Cruiser Hyryder.

Plug-in Hybrid (PHEV)

Function: PHEVs have a larger battery and more powerful motor, allowing for a significant electric-only range (sometimes 30-60+ miles) before the gas engine is needed.

Charging: The battery can be recharged by plugging into an external powersource (home outlet or public charging station) in addition to regenerative braking and the engine.

Examples: The Volvo XC60 Recharge, Audi A4 35 TFSI e, and Mercedes-Benz C-Class PHEV